

Ten Layers of the Exterior Support Route

Exact Factor Ledger, Closed Branches, and the Proof Frontier

Prime Dynamics Theory Program

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Abstract

We audit the ten-paper sequence RH-110–RH-119 as one proof dependency system. Six layers are constructive, two rigorously close tempting routes, and two synthesize the result. The exact ledger is centered on $q_4 = \nu_4/\Lambda_{23}$, with trace–concentration and directional-Rayleigh variants for the normalized fourth volume. Four branches are now explicitly closed: global norm-only wedge perturbation as an improvement, arbitrary fixed-frame universal capture, unguarded fusion of independently rounded Gram assemblies, and finite-anchor fitting as an all-level proof. The five-scale archive is complete for its own finite problem: all 322 genuinely supported labelled records receive an internally consistent adaptive certificate. An AND/OR audit nevertheless shows that eventual all-level support is not reachable from the proved nodes. Its inclusion-minimal mathematical frontier is a three-way antichain: a direct-margin packet, a trace–concentration packet, or a directional-Rayleigh packet. Adding any one completes the conditional graph; validated numerical closure also requires all-level outward admissibility. None of these three physical packets is presently proved. We therefore prioritize a cross-level directional-Rayleigh transfer inequality and make no Hilbert–Polya, zero, or Riemann Hypothesis claim.

1 Scope of the review

The purpose of RH-119 is not to turn a long finite computation into an asymptotic theorem. It is to state exactly what the last ten layers proved, which branches were eliminated, and where every remaining proof path crosses an unproved physical input. The distinction matters because RH-117 proves that finite anchors can match smooth bounded continuations with incompatible zero-scale behavior.

We use three status classes. A *constructive* layer adds a valid bound or algorithm. A *negative* layer proves that a proposed information class cannot improve or cannot determine the target. A *synthesis* layer assembles already valid pieces and audits their logical dependencies. Negative results are part of the route: they prevent the same false shortcut from reappearing under different notation.

2 Exact factor ledger

For a nonzero projected cross K , write

$$q_j = \frac{s_j(K)}{s_1(K)}, \quad \nu_4 = q_2 q_3 q_4, \quad \Lambda_{23} = q_2 q_3.$$

The central identity is written without any asymptotic interpretation:

$$q_4 = \frac{\nu_4}{\Lambda_{23}}. \tag{1}$$

Consequently, a valid volume lower V^- and capacity upper L^+ give $q_4 \geq V^-/L^+$.

The trace variant defines

$$\Theta_4 = \frac{e_4(s(K)^2)}{s_1(K)^8}, \quad \kappa_4 = \frac{e_4(s(K)^2)}{s_1(K)^2 s_2(K)^2 s_3(K)^2 s_4(K)^2},$$

and has the exact identity

$$\nu_4^2 = \frac{\Theta_4}{\kappa_4}. \tag{2}$$

Thus trace lower, concentration upper, and capacity upper produce

$$q_4 \geq \frac{\sqrt{\Theta_4^-/\kappa_4^+}}{L^+}.$$

The directional variant uses a recent four-frame action with normalized volume lower a^- and a relative residual-Gram factor γ^+ . The Rayleigh comparison from RH-114 gives

$$q_4 \geq \frac{(1 - \gamma^+)_+^4 a^-}{L^+}.$$

All three statements are ordinary one-sided singular-value arguments [Horn and Johnson, 1991]. Their numerical fusion is valid only after every factor encloses the same K , or after explicit outward transport [Moore, 1966]. The information-optimal fusion, absent further cross-candidate relations, is the maximum of the admitted lower bounds.

This ledger is the invariant core of RH-110–RH-119. It says nothing by itself about primes, zeta zeros, or an operator spectrum; those require separate bridges.

3 Layer-by-layer ledger

The six constructive layers solve algebraic, perturbative, directional, compositional, or algorithmic subproblems. They do not jointly prove a physical inequality across all later scales. The two negative layers are sharp: RH-112 proves universal domination of global wedge-Lipschitz by product Weyl, while RH-117 constructs bounded smooth continuations through the same anchors with limits 0, an interior constant, and 1.

RH-115 contributes both kinds of knowledge. Its admitted composite is valid on all 360 records, but an exact-directional diagnostic assembled by a second binary64 path exceeds the reference on one weak physical record. The outward-admissibility filter therefore closes unguarded cross-assembly fusion as a proof method. RH-116 resolves the finite instance by using one assembly path and proves that first passage in memory depth is cost-minimal.

layer	class	principal result	route consequence
110	constructive	capacity factorization and enclosure	factorized route opened
111	constructive	trace–concentration certificate	physical trace route opened
112	negative	sharp global wedge barrier	norm-only improvement closed
113	constructive	directional frame variation	structured route retained
114	constructive	PSD Rayleigh tail bound	relative gamma isolated
115	constructive	composite and outward filter	unsafe fusion closed
116	constructive	monotone optimal memory depth	finite search completed
117	negative	finite-anchor nonidentifiability	extrapolation closed
118	synthesis	conditional liminf theorem	physical packets isolated
119	synthesis	proof-frontier audit	roadmap revised

Table 1: The ten-layer status ledger.

4 Closed branches

The review records four explicit closures.

1. **Global norm-only wedge improvement.** The sharp global perturbation formula is always dominated by the product-Weyl lower, so it cannot improve that certificate class.
2. **Arbitrary fixed-frame universal capture.** RH-113 gives an exact blindness example. Directional arguments require an adapted frame or a separate capture hypothesis.
3. **Unguarded cross-assembly fusion.** Floating-point agreement is not an inequality direction. A common enclosed operator or an outward transport loss is mandatory.
4. **Finite-anchor asymptotic fitting.** Any finite capacity anchors are compatible with incompatible bounded smooth limits. A fit is evidence, not an all-level implication.

These closures do not invalidate the broad exterior route. They remove shortcuts that lack enough information.

5 The proof-frontier antichain

We formalize the current route as a monotone AND/OR proof graph. The proved nodes include the factor identities, trace and directional conversions, monotone maximum, nested-depth first passage, outward-degradation rule, RH-118 conditional liminf theorem, finite archive support, and RH-117 finite-anchor nonidentifiability.

Let P_D , P_T , and P_R denote the all-level direct-margin, trace–concentration, and directional-Rayleigh packets. Let C denote the proved conditional liminf theorem and let E denote eventual fourth-mode support. The only current rules producing E are

$$(C \wedge P_D) \Rightarrow E, qquad (C \wedge P_T) \Rightarrow E, qquad (C \wedge P_R) \Rightarrow E. \quad (3)$$

Theorem 1 (Proof-frontier antichain). *Relative to the audited rule system (3), eventual support is not reachable from the current proved nodes. Its inclusion-minimal missing sets are exactly*

$$\{P_D\}, \quad \{P_T\}, \quad \{P_R\}.$$

Adding any one packet makes E reachable. If validated closure through independently assembled numerical paths is required, each minimal set also contains all-level outward admissibility.

Proof. C is proved, while none of P_D, P_T, P_R is proved. Therefore no antecedent in (3) is currently complete. Supplying any one packet completes its corresponding antecedent. No proper subset of a singleton can do so, and the three singleton sets do not contain one another. Validated numerical closure is an additional conjunction with outward admissibility, which adds that node to each minimal set. $\square$

This is a theorem about the present proof ledger, not a theorem asserting that these are the only mathematical ideas possible. A future paper may add a genuinely new rule. The review makes such an addition explicit rather than allowing it to enter as an unstated assumption.

The complete finite audit does not produce an extra rule to E . RH-118 records 322 genuinely supported labelled updates and RH-116 certifies all 322, but RH-117 proves why finite support cannot imply the missing liminf packet.

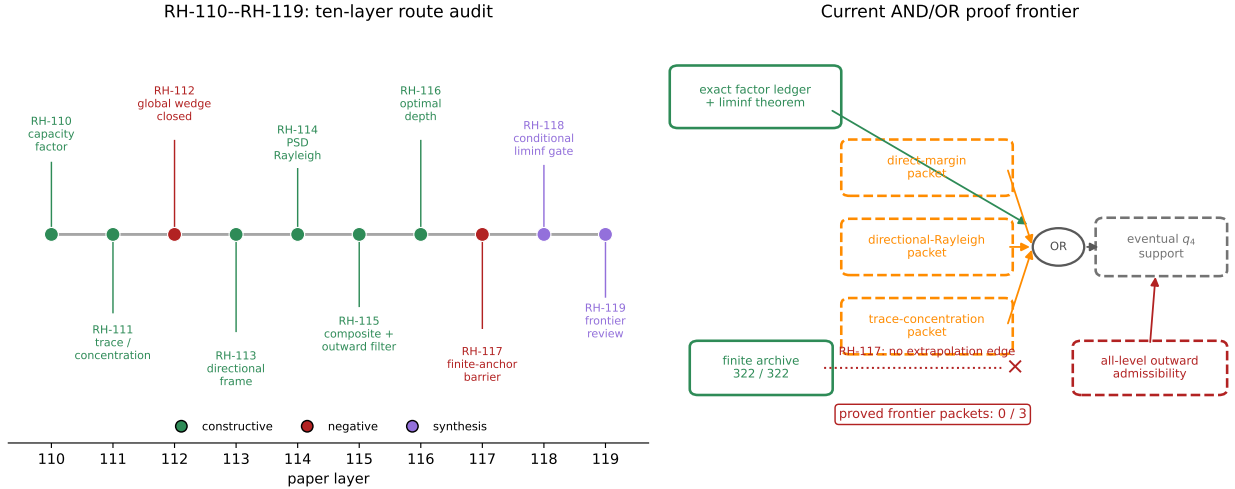


Figure 1: Left: the ten-layer sequence. Right: the proved nodes, three open physical packets, and the absent finite-to-asymptotic edge.

6 Revised roadmap

The graph-theoretically shortest route is P_D , one compound direct-margin inequality. It is also close to restating the desired support conclusion, so proving it may offer little mechanism. The strongest next research target is instead P_R , the directional-Rayleigh packet:

1. derive a cross-level transfer inequality for normalized recent-frame volume and the relative tail Gramian;
2. prove a scale-uniform bound $\gamma^+ < 1$ with enough quantitative margin after division by capacity;
3. use one assembly path from the start, or attach an outward transport radius to every independent path;
4. audit the resulting recurrence on the existing five anchors, while keeping the proof independent of those anchors;
5. pursue a direct-margin recurrence in parallel as the shortest fallback.

The trace-concentration packet remains mathematically valid but is third priority: it never wins the RH-115 finite maximum, and it lacks an independent physical exterior-trace source law. That route should be reactivated if a new source-level inequality appears.

Only after one all-level packet is proved should the program reconsider the next downstream bridge. Nothing in RH-110–RH-119 constructs a self-adjoint Hilbert–Polya operator, identifies a Riemann zeta zero as an eigenvalue, or proves the Riemann Hypothesis.

7 Conclusion

The route has not reached a dead end, but its remaining difficulty is no longer hidden. Exact algebra, finite perturbation, directional comparison, certificate fusion, and memory optimization are in place. Four false paths are marked closed. The next advance must be a genuine cross-level physical inequality for at least one frontier packet, with outward numerical logic built in from the beginning.

References

- Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.
- Ramon E. Moore. *Interval Analysis*. Prentice-Hall, 1966.